

6.2. DIVERGENCE THEOREM OF GAUSS

(Relationship b/w surface and volume integrals)

→ If $\vec{F}$ is a vector point function having continuous first order P.D.E.s in the region V bounded by a closed surface S , then

$$\iiint_V \nabla \cdot \vec{F} \, dV = \iint_S \vec{F} \cdot \hat{n} \, dS,$$

$\hat{n}$ is the unit outwards drawn unit normal vector to the surface S .

Q. For any closed surface S , P.T. $\iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS = 0$.

By the div theorem,

$$\begin{aligned} \iint_S \text{curl } \vec{F} \cdot \hat{n} \, dS &= \iiint_V \text{div}(\text{curl } \vec{F}) \, dV \\ &= \iiint_V 0 \, dV \\ &= 0 \end{aligned}$$

$$+ j(6+50) - i(32+32) = 4 = 0$$

Q. Evaluate $\iint_S \vec{r} \cdot \hat{n} \, dS$, where S is a closed surface.

By div thm,

$$\iint_S \vec{r} \cdot \hat{n} \, dS = \iiint_V \nabla \cdot \vec{r} \, dV = \iiint_V 3 \, dV$$

$$= 3V$$

Q. Use div. thm to S.T. $\oint_S \nabla r^2 d\vec{s} = 6V$, where S is any closed surface enclosing a volume V ?

By div thm,

$$\oint_S \nabla r^2 d\vec{s} = \int_V \text{div} (\nabla r^2) dV$$

$$= \int_V \nabla \cdot (\nabla r^2) dV$$

$$= \int_V \nabla^2 r^2 dV$$

$$= \int_V \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) (x^2 + y^2 + z^2) dV$$

$$= \int_V (2+2+2) dV$$

$$= \int_V 6 dV = \underline{6V}$$

Ans

Q. Find $\iint_S \vec{F} \cdot \hat{n} ds$, where $\vec{F} = (2x+3z)\hat{i} - (x^2+y)\hat{j} + (y^2+2z)\hat{k}$, where S is the surface of sphere having radius 3 and centre $(3, -1, 2)$?

Let V be the volume of the sphere.

By div thm,

$$\iint_S \vec{F} \cdot \hat{n} ds = \iiint_V \text{div} \vec{F} dV$$

$$= \iiint_V \left[\frac{\partial}{\partial x} (3x+3z) + \frac{\partial}{\partial y} (-x^2-y) + \frac{\partial}{\partial z} (y^2+3z) \right] dV$$

$$= \iiint_V (2-1+2) dV$$

$$= 3 \iiint_V dV = 3V$$

Here $V = \frac{4}{3} \pi r^3$

$$= \frac{4}{3} \times \pi (3)^3 = 36\pi$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} dS = 3 \times 36\pi = 108\pi$$

Q. Verify div thm for $\vec{F} = (x^2-yz)\hat{i} + (y^2-zx)\hat{j} + (z^2-xy)\hat{k}$ taken over the rectangular parallelepiped, $0 \leq x \leq a$, $0 \leq y \leq b$, $0 \leq z \leq c$?

$$\begin{aligned} \text{div } \vec{F} &= \frac{\partial}{\partial x} (x^2-yz) + \frac{\partial}{\partial y} (y^2-zx) + \frac{\partial}{\partial z} (z^2-xy) \\ &= 2(x+y+z) \end{aligned}$$

$$\therefore \iiint_V \text{div } \vec{F} dV = \int_0^c \int_0^b \int_0^a 2(x+y+z) dx dy dz$$

$$= 2 \int_0^c \int_0^b \left(\frac{x^2}{2} + yx + zx \right) \Big|_0^a dy dz$$

$$= 2 \int_0^c \int_0^b \left(\frac{a^2}{2} + ya + za \right) dy dz$$

$$= 2 \int_0^c \left(\frac{a^2}{2} b + \frac{y^2}{2} a + zay \right) \Big|_0^b dz$$

$$\begin{aligned}
 &= 2 \int_0^c \left(\frac{a^2 b}{2} + \frac{ab^2}{2} + abz \right) dz \\
 &= 2 \left(\frac{a^2 b}{2} z + \frac{ab^2}{2} z + \frac{abz^2}{2} \right)_0^c \\
 &= 2 \left(\frac{a^2 bc}{2} + \frac{ab^2 c}{2} + \frac{abc^2}{2} \right) \\
 &= a^2 bc + ab^2 c + abc^2 \\
 &= \underline{\underline{abc(a+b+c)}} \quad \text{--- (1)}
 \end{aligned}$$

To evaluate surface integral

Divide the closed surface S of the rectangular parallelepiped into 6 parts.

S_1 :- the face $OACB$

S_2 :- the face $CB'PA'$

S_3 :- " " face $OBA'C$

S_4 :- " " face $AC'PB'$

S_5 :- " " face $OCB'A'$

S_6 :- " " face $B'A'PC'$

On S_1

$$z=0, \quad \hat{n} = -\hat{k}, \quad \vec{F} = x^2 \hat{i} + y^2 \hat{j} - xy \hat{k}$$

$$\therefore \vec{F} \cdot \hat{n} = xy$$

$$\therefore \iint_{S_1} \vec{F} \cdot \hat{n} \, ds = \int_0^b \int_0^a xy \, dx \, dy = \frac{a^2 b^2}{4}$$

On S_2

$$z=c, \quad \hat{n} = \hat{k}, \quad \vec{F} = (x^2 - cy) \hat{i} + (y^2 - cx) \hat{j} + (c^2 - xy) \hat{k}$$

$$\iint_{S_1} \vec{F} \cdot \hat{n} dS = \int_0^b \int_0^a (c^2 - xy) dx dy = abc^2 - \frac{a^2 b^2}{4}$$

On S_2
 $x=0, \hat{n} = -\hat{i}, \vec{F} = -yz\hat{i} + y^2\hat{j} + z^2\hat{k}$

$$\vec{F} \cdot \hat{n} = -yz$$

$$\iint_{S_3} \vec{F} \cdot \hat{n} dS = \int_0^c \int_0^b yz dy dz = \frac{b^2 c^2}{4}$$

On S_4

$x=a, \hat{n} = \hat{i}, \vec{F} = (a^2 - yz)\hat{i} + (y^2 - az)\hat{j} + (z^2 - ay)\hat{k}$

$$\vec{F} \cdot \hat{n} = (a^2 - yz)$$

$$\therefore \iint_{S_4} \vec{F} \cdot \hat{n} dS = \int_0^c \int_0^b (a^2 - yz) dy dz$$

$$= abc^2 - \frac{b^2 c^2}{4}$$

On S_5

$y=0, \hat{n} = -\hat{j}, \vec{F} = x^2\hat{i} - zm\hat{j} + z^2\hat{k}$

$$\vec{F} \cdot \hat{n} = -zm$$

$$\iint_{S_5} \vec{F} \cdot \hat{n} dS = \int_0^a \int_0^c -zm dz dm = -\frac{a^2 c^2}{4}$$

On S_6

$y=b, \hat{n} = \hat{j}, \vec{F} = (x^2 - bz)\hat{i} + (b^2 - zm)\hat{j} + (z^2 - bx)\hat{k}$

$$\vec{F} \cdot \hat{n} = (a^2 - bz)\hat{i} + (b^2 - az)\hat{j} + (c^2 - bx)\hat{k} = b^2 - az$$

$$\iint_{S_4} \vec{F} \cdot \hat{n} \, ds = \int_0^a \int_0^b (b^2 - az) \, dz \, dx = ab^2 - \frac{a^2 c^2}{4}$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} \vec{F} \cdot \hat{n} \, ds + \iint_{S_2} \vec{F} \cdot \hat{n} \, ds + \iint_{S_3} \vec{F} \cdot \hat{n} \, ds +$$

$$\iint_{S_4} \vec{F} \cdot \hat{n} \, ds + \iint_{S_5} \vec{F} \cdot \hat{n} \, ds + \iint_{S_6} \vec{F} \cdot \hat{n} \, ds$$

$$= \frac{a^2 b^2}{4} + abc^2 + \frac{a^2 b^2}{4} + \frac{b^2 c^2}{4} + a^2 bc - \frac{b^2 c^2}{4} + \frac{a^2 c^2}{4} +$$

$$ab^2c - \frac{a^2 c^2}{4}$$

$$= abc^2 + a^2 bc + ab^2c = abc(a+b+c) \quad \text{--- (2)}$$

$$\therefore \textcircled{1} = \textcircled{2} \Rightarrow \text{div thm.}$$

HW Q. Verify div thm for $\vec{F} = 4xy\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the region bdd by the cylinder $x^2 + y^2 = 4$, $z=0$, $z=3$.

$$\text{div } \vec{F} = 4 - 4y + 2z$$

$$\therefore \iiint_V \text{div } \vec{F} \, dV = \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^3 (4 - 4y + 2z) \, dz \, dy \, dx$$

$$= 84\pi \quad \text{--- (1)}$$

To evaluate surface integral.

Divide S into 3 parts.

S_1 : circular base in the plane $z=0$.

S_2 : " " " " $z=3$

S_3 : circular surface of the cylinder, given by the equation $x^2 + y^2 = 4$.

On S_1

$$z=0, \hat{n} = -\hat{k}, \vec{F} = 4x\hat{i} - 2y^2\hat{j} \quad \leftarrow \text{①}$$

$$\vec{F} \cdot \hat{n} = 0$$

$$\iint_{S_1} \vec{F} \cdot \hat{n} dS = 0.$$

On S_2

$$z=3, \hat{n} = \hat{k}, \vec{F} = 4x\hat{i} - 2y^2\hat{j} + 9\hat{k}$$

$$\begin{aligned} \iint_{S_2} \vec{F} \cdot \hat{n} dS &= \iint_{S_2} 9 dx dy = 9 \times \text{area of } S_2 \\ &= 9 \times \pi \times 2^2 = \underline{\underline{36\pi}}. \end{aligned}$$

On S_3

$$x^2 + y^2 = 4.$$

$$\text{vector normal to } S_3 = \nabla (x^2 + y^2) = 2x\hat{i} + 2y\hat{j}$$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j}}{\sqrt{4x^2 + 4y^2}} = \frac{x\hat{i} + y\hat{j}}{2}$$

$$\vec{F} \cdot \hat{n} = 2x^2 - y^3$$

Let $x = 2 \cos \theta$, $y = 2 \sin \theta$

$ds = 2 d\theta dz$, $0 \leq z \leq 3$, $0 \leq \theta \leq 2\pi$

$$\iint_S \vec{F} \cdot \hat{n} dS = \int_0^{2\pi} \int_0^3 [2(2 \cos \theta)^2 - (2 \sin \theta)^3] 2 dz d\theta$$

$$= 48\pi$$

$$\therefore \iint_S \vec{F} \cdot \hat{n} dS = 0 + 36\pi + 48\pi = 84\pi \quad \text{--- (2)}$$

① = ② $\Rightarrow$ div thm

Q. Evaluate $\iint_S (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}) \cdot \hat{n} dS$, where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ above the xy -plane and bdd by this plane?

Let V be the volume enclosed by the surface S .
Then by divergence thm,

$$\iint_S (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}) \cdot \hat{n} dS = \iiint_V \text{div} (y^2 z^2 \hat{i} + z^2 x^2 \hat{j} + z^2 y^2 \hat{k}) dV$$

$$= \iiint_V \left[\frac{\partial}{\partial x} (y^2 z^2) + \frac{\partial}{\partial y} (z^2 x^2) + \frac{\partial}{\partial z} (z^2 y^2) \right] dV$$

$$= \iiint_V 2zy^2 dV = 2 \iiint_V zy^2 dV$$

Changing to spherical polar coordinates,

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

$$dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq \pi/2, \quad 0 \leq \phi \leq 2\pi$$

$$\therefore 2 \iiint_V z y^2 \, dV = 2 \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 (r \cos \theta) (r^2 \sin^2 \theta \sin^2 \phi) r^2 \sin \theta \, dr \, d\theta \, d\phi$$

$$= 2 \int_0^{2\pi} \int_0^{\pi/2} \int_0^1 r^5 \cos \theta \sin^3 \theta \sin^2 \phi \, dr \, d\theta \, d\phi$$

$$= 2 \int_0^{2\pi} \int_0^{\pi/2} \left[\frac{r^6}{6} \right]_0^1 \cos \theta \sin^3 \theta \sin^2 \phi \, d\theta \, d\phi$$

$$= \frac{1}{3} \int_0^{2\pi} \int_0^{\pi/2} \cos \theta \sin^3 \theta \sin^2 \phi \, d\theta \, d\phi$$

$\frac{64}{4}$

Q. Evaluate $\iint_S \vec{F} \cdot \hat{n} \, ds$ over the entire surface of the region above the xy -plane bounded by the cone $z^2 = x^2 + y^2$ and the plane $z=4$ if $\vec{F} = 4xz\mathbf{i} + xyz^2\mathbf{j} + 3z\mathbf{k}$

If V is the volume enclosed by S , then V is bdd by the surfaces $z=0, z=4, z^2=x^2+y^2$.

By div thm,

$$\begin{aligned} \iint_S \vec{F} \cdot \hat{n} \, dS &= \iiint_V \operatorname{div} \vec{F} \, dV \\ &= \iiint_V \left[\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(xyz^2) + \frac{\partial}{\partial z}(3z^2) \right] dV \end{aligned}$$

$$\begin{aligned} &= \iiint_V (4z + xz^2 + 3) \, dV \\ &= \int_0^4 \int_{-z}^z \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} (4z + xz^2 + 3) \, dx \, dy \, dz \end{aligned}$$

$$= 2 \int_0^4 \int_{-z}^z \int_0^{\sqrt{z^2-y^2}} (4z + \cancel{xz^2} + 3) \, dx \, dy \, dz, \quad \left(\because \int_{-\sqrt{z^2-y^2}}^{\sqrt{z^2-y^2}} x \, dx = 0 \right)$$

$$= 2 \int_0^4 \int_{-z}^z (\sqrt{z^2-y^2}) (4z+3) \, dy \, dz$$

$$= 4 \int_0^4 \int_0^z \sqrt{z^2-y^2} (4z+3) \, dy \, dz$$

$$= 4 \int_0^4 \left[\frac{y}{2} \sqrt{z^2-y^2} + \frac{z^2}{2} \sin^{-1} \frac{y}{z} \right]_0^z (4z+3) \, dz$$

$$= 4 \int_0^4 (4z+3) \left[\frac{z^2}{2} \sin^{-1}(1) \right] dz = 4 \times \frac{\pi}{4} \int_0^4 \frac{z^2}{2} (4z+3) \, dz$$

$$= \frac{\pi}{2} \int_0^4 (4z^3 + 3z^2) \, dz$$

$$= \pi \left[z^4 + 2^3 \right]_0^4$$

$$= \pi (256 + 64) = \underline{\underline{320\pi}}$$

Q. Evaluate $\iint_S (\nabla \times \vec{A}) \cdot \hat{n} \, dS$, where S is the surface of the cone $z = 2 - \sqrt{x^2 + y^2}$ above the xy -plane and $\vec{A} = (x-z)\hat{i} + (x^2 + yz)\hat{j} - 3xy^2\hat{k}$?

The surface $z = 2 - \sqrt{x^2 + y^2}$ meets the xy -plane in a circle c given by $x^2 + y^2 = 4$, $z = 0$. Let S_1 be the plane region bdd by the circle c . If S' is the surface consisting of the surface S and S_1 , then S' is a closed surface.

$\therefore$ By div thm,

$$\iiint_{S'} \text{curl } \vec{A} \cdot \hat{n} \, dS = 0$$

$$\Rightarrow \iint_S \text{curl } \vec{A} \cdot \hat{n} \, dS + \iint_{S_1} \text{curl } \vec{A} \cdot \hat{n} \, dS = 0$$

$$\Rightarrow \iint_S \text{curl } \vec{A} \cdot \hat{n} \, dS - \iint_{S_1} \text{curl } \vec{A} \cdot \hat{k} \, dS = 0 \quad (\text{on } S_1, \hat{n} = -\hat{k})$$

$$\Rightarrow \iint_S \text{curl } \vec{A} \cdot \hat{n} \, dS = \iint_{S_1} \text{curl } \vec{A} \cdot \hat{k} \, dS$$

$$\text{Now, } \text{curl } \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x-z & x^2+yz & -3xy^2 \end{vmatrix}$$

$$= \hat{i}(-6xy - y) + \hat{j}(-1 + 3y^2) + \hat{k}(3x^2)$$

$$\therefore \text{curl } \vec{A} \cdot \hat{K} = 3r^2.$$

$$\therefore \iint_S \text{curl } \vec{A} \cdot \hat{n} \, dS = \iint_S 3r^2 \, dS$$

$$= \int_0^{2\pi} \int_0^2 3(r^2 \cos^2 \theta) \cdot r \, dr \, d\theta$$

$$= 3 \int_0^{2\pi} \int_0^2 r^3 \cos^2 \theta \, dr \, d\theta$$

$$= 3 \int_0^{2\pi} \left[\frac{r^4}{4} \right]_0^2 \cos^2 \theta \, d\theta$$

$$= \frac{3}{4} \int_0^{2\pi} 2^4 \left(\frac{1 + \cos 2\theta}{2} \right) d\theta$$

$$= \frac{3 \cdot 4 \cdot 4^2}{4 \cdot 2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= 6 [2\pi - 0] = \underline{\underline{12\pi}}$$